

IDENTIFYING INFORMATION:

NAME: Matthis, Jonathan Samir

ORCID iD: <https://orcid.org/0000-0003-3683-646X>

POSITION TITLE: President

PRIMARY ORGANIZATION AND LOCATION: FreeMoCap Foundation, Inc. , Boston, Massachusetts, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Texas at Austin, Austin, Texas, United States	Postdoctoral Fellow	09/2014 - 06/2019	Perceptual Systems
Rensselaer Polytechnic Institute, Troy, New York, United States	PHD	05/2011 - 05/2014	Cognitive Science
Rensselaer Polytechnic Institute, Troy, New York, United States	MS	09/2008 - 05/2011	Cognitive Science
University of Maryland Baltimore County, Baltimore, MD, United States	BA	09/2005 - 05/2007	Philosophy

Appointments and Positions

2021 - present President, FreeMoCap Foundation, Inc. , Boston, Massachusetts, United States

2021 - present President/CEO, FreeMoCap Foundation, Inc., Boston, Massachusetts, United States

2019 - 2026 Assistant Professor, Northeastern University, Boston, Massachusetts, United States

Products**Products Most Closely Related to the Proposed Project**

1. Matthis JS, Yates JL, Hayhoe MM. Gaze and the Control of Foot Placement When Walking in Natural Terrain. Curr Biol. 2018 Apr 23;28(8):1224-1233.e5. PubMed Central PMCID: [PMC5937949](https://pubmed.ncbi.nlm.nih.gov/PMC5937949/).
2. Matthis JS, Barton SL, Fajen BR. The critical phase for visual control of human walking over complex terrain. Proc Natl Acad Sci U S A. 2017 Aug 8;114(32):E6720-E6729. PubMed Central PMCID: [PMC5558990](https://pubmed.ncbi.nlm.nih.gov/PMC5558990/).
3. Jonathan Samir Matthis, Philip Queen, Aaron Cherian, Trent Wirth, Endurance Idehen. FreeMoCap: A free, open source markerless motion capture system. 2026 August. Available from: <https://zenodo.org/doi/10.5281/zenodo.22131400> DOI: 10.5281/ZENODO.22131400
4. jonmatthis, Endurance Idehen, philipqueen, Mac Prible, Aaron Cherian. SkellyCam: The camera backend for the FreeMoCap software. 2026 April. Available from:

<https://zenodo.org/doi/10.5281/zenodo.19921107> DOI: 10.5281/ZENODO.19921107

5. Matthis JS, Muller KS, Bonnen KL, Hayhoe MM. Retinal optic flow during natural locomotion. PLoS Comput Biol. 2022 Feb;18(2):e1009575. PubMed Central PMCID: [PMC8896712](https://pubmed.ncbi.nlm.nih.gov/PMC8896712/).

Other Significant Products, Whether or Not Related to the Proposed Project

1. Matthis JS, Barton SL, Fajen BR. The biomechanics of walking shape the use of visual information during locomotion over complex terrain. J Vis. 2015 Mar 18;15(3) PubMed Central PMCID: [PMC4365914](https://pubmed.ncbi.nlm.nih.gov/PMC4365914/).
2. Matthis JS, Fajen BR. Visual control of foot placement when walking over complex terrain. J Exp Psychol Hum Percept Perform. 2014 Feb;40(1):106-15. PubMed PMID: [23750964](https://pubmed.ncbi.nlm.nih.gov/23750964/); NIHMSID: NIHMS1050949.
3. philipqueen, jonmatthis, Aaron Cherian. SkellyClicker: For labeling and traing multi-camera video datasets. 2026 April. Available from: <https://zenodo.org/doi/10.5281/zenodo.19389889> DOI: 10.5281/ZENODO.19389889
4. philipqueen, jonmatthis. SkellyTracker: The image analysis backend for the FreeMoCap Project. 2023 December. Available from: <https://zenodo.org/doi/10.5281/zenodo.10395641> DOI: 10.5281/ZENODO.10395641
5. Aaron Cherian, jonmatthis. freemocap/skellyforge: 2023.12.1005. 2023 December. Available from: <https://zenodo.org/doi/10.5281/zenodo.10395649> DOI: 10.5281/ZENODO.10395649

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Matthis, Jonathan Samir in SciENCv on 2026-08-28 15:03:59